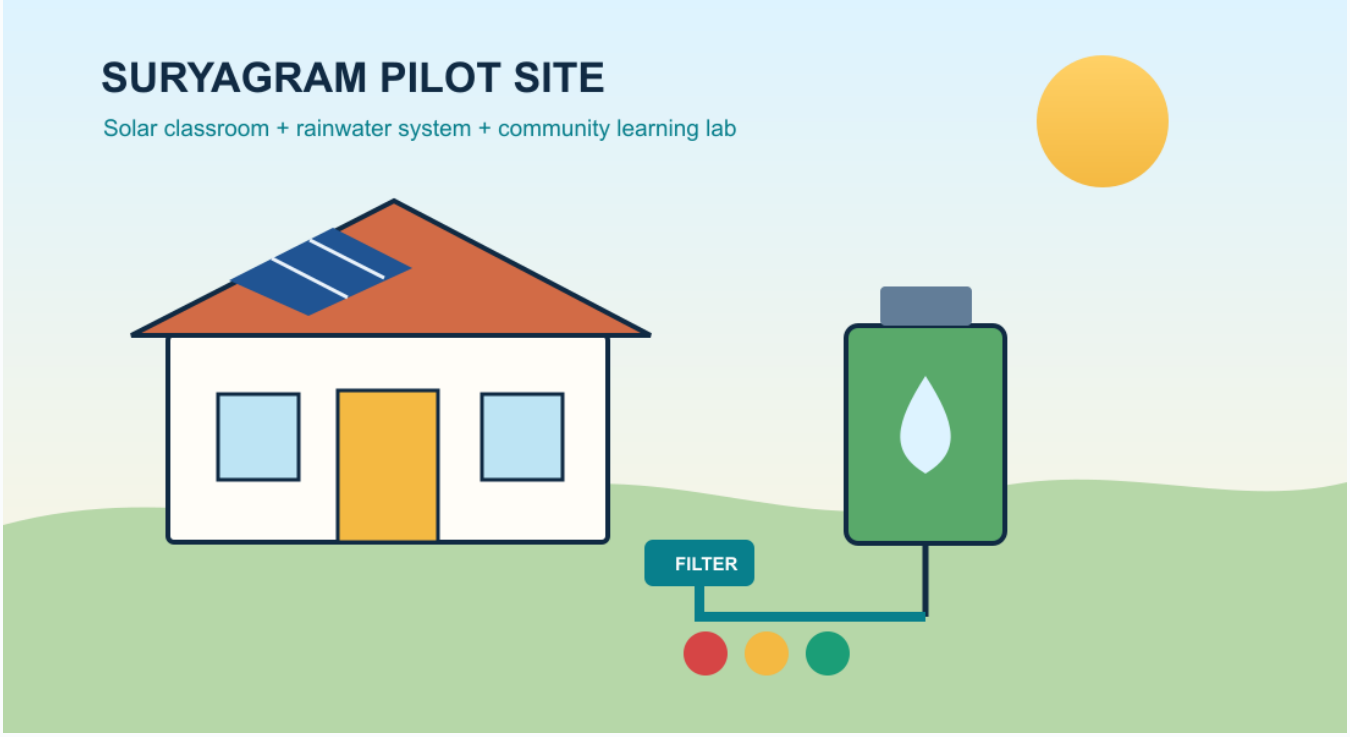
20-PAGE MULTIMODAL TEST PACK

Suryagram Smart Learning & Water Resilience Pilot

Annual Impact Report 2025



Created for Aiwana ingestion, retrieval, visual-evidence and Hinglish QA

Fictional report: all names, sites and measurements are synthetic test data.

120	18 kW	50,000 L
students enrolled	solar capacity	water storage

How to Use This Document in Aiwana

Purpose: this report is designed to test grounded chat over narrative text, exact numbers, tables, charts, diagrams, illustrations, comparisons, multilingual queries and scenario reasoning.

RECOMMENDED TEST FLOW

1. **Upload**

   Add this PDF to a new Knowledge Space and wait until indexing is ready.

2. **Ask facts**

   Ask for capacity, dates, totals and programme results; validate page citations.

3. **Ask visuals**

   Use Explain Visual mode on the energy chart, site plan or field gallery.

4. **Ask Hinglish**

   Try: Paani availability ka trend simple language mein samjhao.

Test modes covered

Answer · Summarize · Explain Visual · Compare · Scenario · Quiz Me · Hinglish · Layman

Document Contents

Pages	What to test
03-05	Programme overview and baseline
06-10	Energy, water, attendance, learning and digital charts
11-13	Process, expenditure and visual field evidence
14-18	Trends, scenarios, language, comparisons and risks
19-20	Evidence inventory and query challenge set

Executive Summary

Suryagram is a fictional rural learning-centre pilot established to test whether integrated solar power, safe-water infrastructure and offline digital learning can improve attendance and learning conditions in a single academic year.

85%
average attendance in 2025

92%
learners with safe water

98.9 MWh
solar generated in 2025

KEY FINDINGS

- Attendance increased from a weighted baseline of 68% to 85%, with Grade 7 recording the strongest improvement.
- Science assessment scores rose from 44 to 70 out of 100; digital literacy rose from 29 to 76.
- Water availability peaked at 770 litres/day in August after monsoon recharge.
- Actual programme spending was INR 51.1 lakh, INR 2.2 lakh below the approved budget.

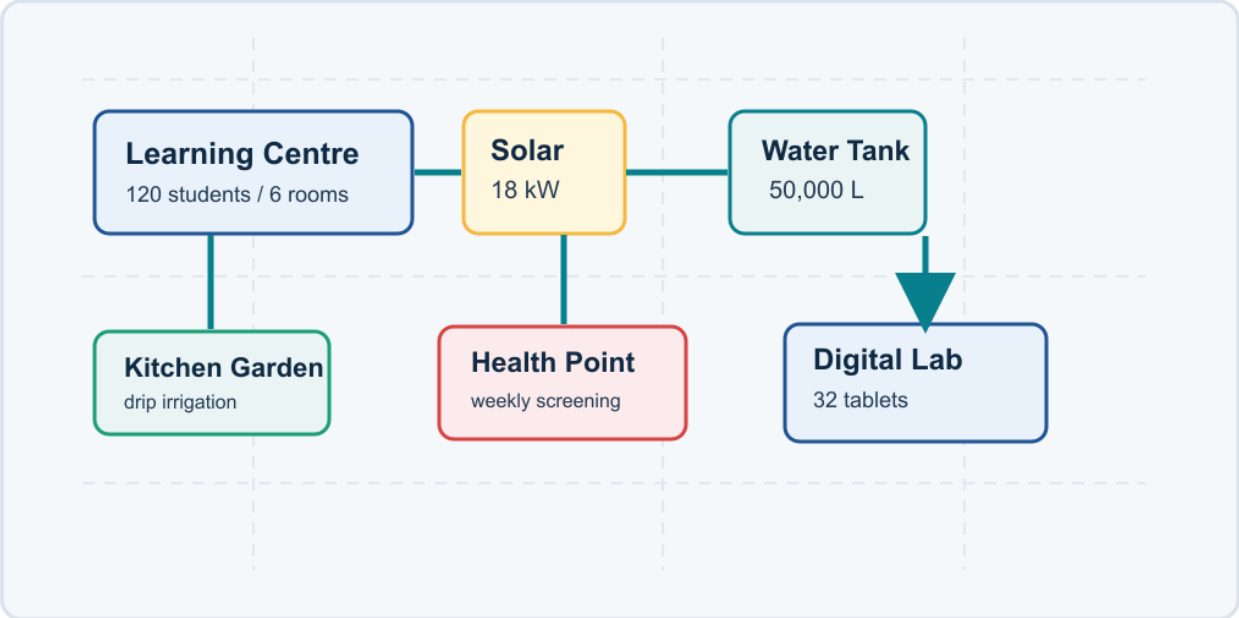
Evidence rule for testing

All conclusions in this report should be traceable to a page, visual or table.
Questions about household income, crop yields, student names or medical diagnoses are intentionally unsupported.

Project Context and Site Plan

The pilot site serves 120 students from Suryagram village. The learning centre combines six classrooms, a 32-tablet digital lab, an 18 kW rooftop solar system, a 50,000-litre rainwater tank, a kitchen garden and a weekly health point.

FIGURE 1. Integrated campus layout and service links



VISUAL TEST

Ask: Which building is connected to both learning and energy functions? Or ask in Hinglish: Tank aur digital lab centre mein kahan placed hain?

Location note

Coordinates and geographic identifiers are omitted by design. Aiwana should explain the illustrated campus layout, but should not claim a real map location.

Baseline Cohort Profile

The same 120 enrolled students were monitored during baseline and endline. Assessment analysis uses 96 students who completed both tests; attendance reporting includes all enrolled students.

Grade	Enrolled	Girls	Boys	Baseline attendance
Grade 6	42	22	20	67%
Grade 7	40	21	19	70%
Grade 8	38	20	18	65%
Total	120	63	57	68%

RETRIEVAL DETAILS

63
girls enrolled

57
boys enrolled

96
matched assessments

Definitions

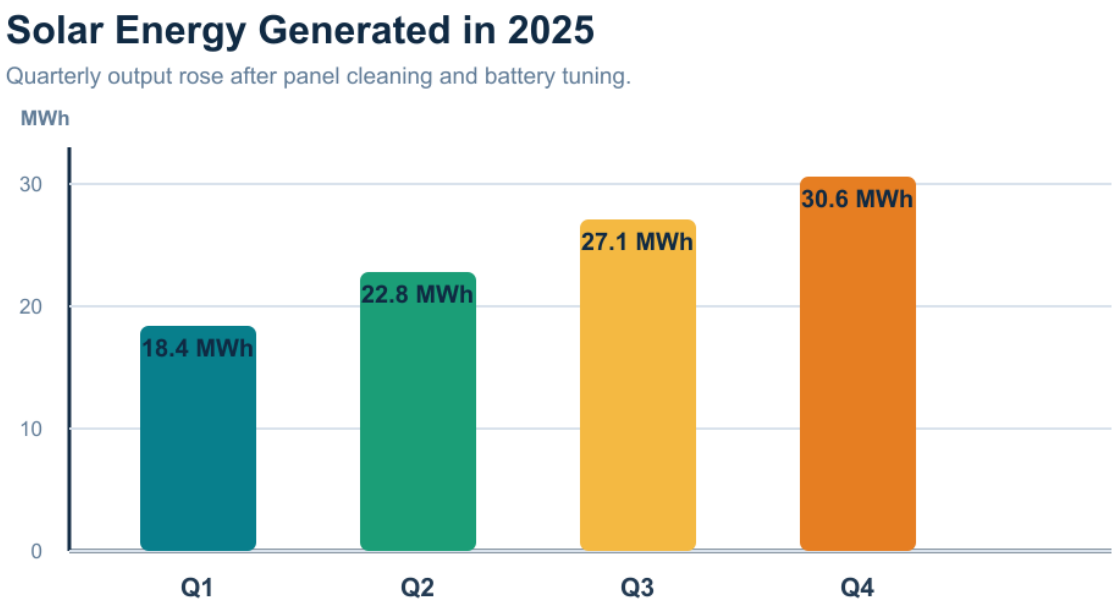
Attendance is the percentage of scheduled school days attended. Safe water access is the percentage of observed school days when treated drinking water was available for all students.

Do not infer

The report contains no caste, disability, household income or individual student performance data.

Energy Performance

The rooftop solar system was commissioned on 10 January 2025. Quarterly generation totalled 98.9 MWh, with Q4 the strongest period at 30.6 MWh after preventive maintenance in September.



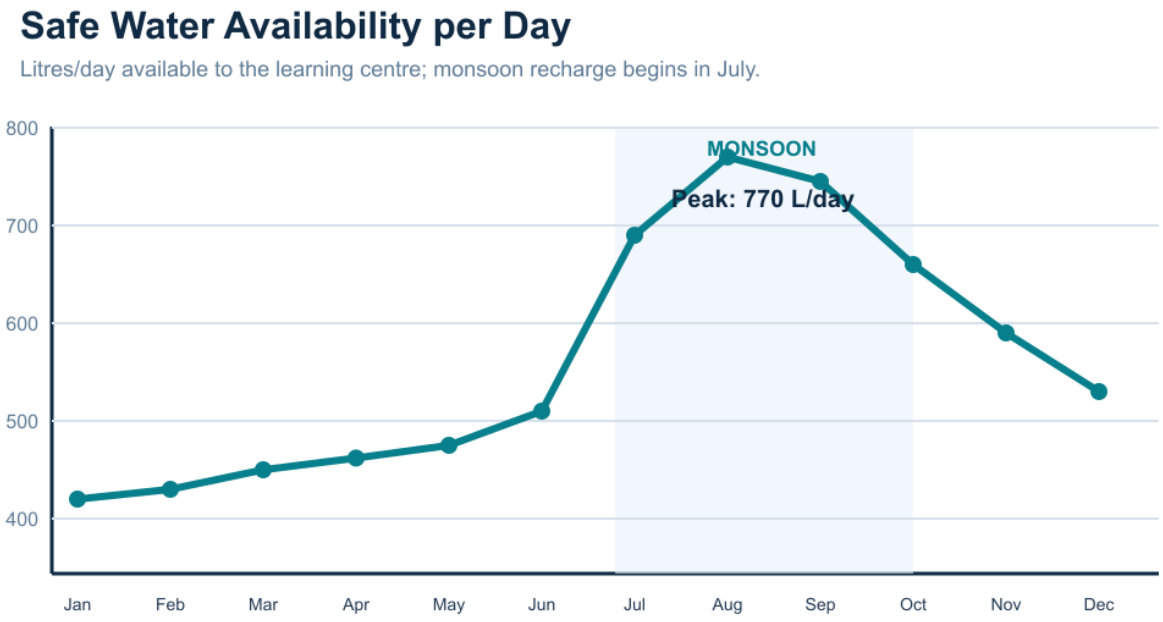
Reading the chart

Q4 generated 12.2 MWh more than Q1. The chart supports a rising quarterly trend; it does not report electricity revenue.

Quarter	Generation (MWh)	Key event
Q1	18.4	Commissioning and calibration
Q2	22.8	Normal operations
Q3	27.1	Panel cleaning completed
Q4	30.6	Battery tuning stabilised supply

Water Security and Seasonal Trend

The rainwater system stores treated water for the learning centre. Daily availability averaged 569 litres across the year and increased sharply during monsoon recharge between July and September.



HINGLISH PROMPT

Paani availability July ke baad kyun badhi, aur peak month kaunsa tha? Simple Hinglish mein batao.

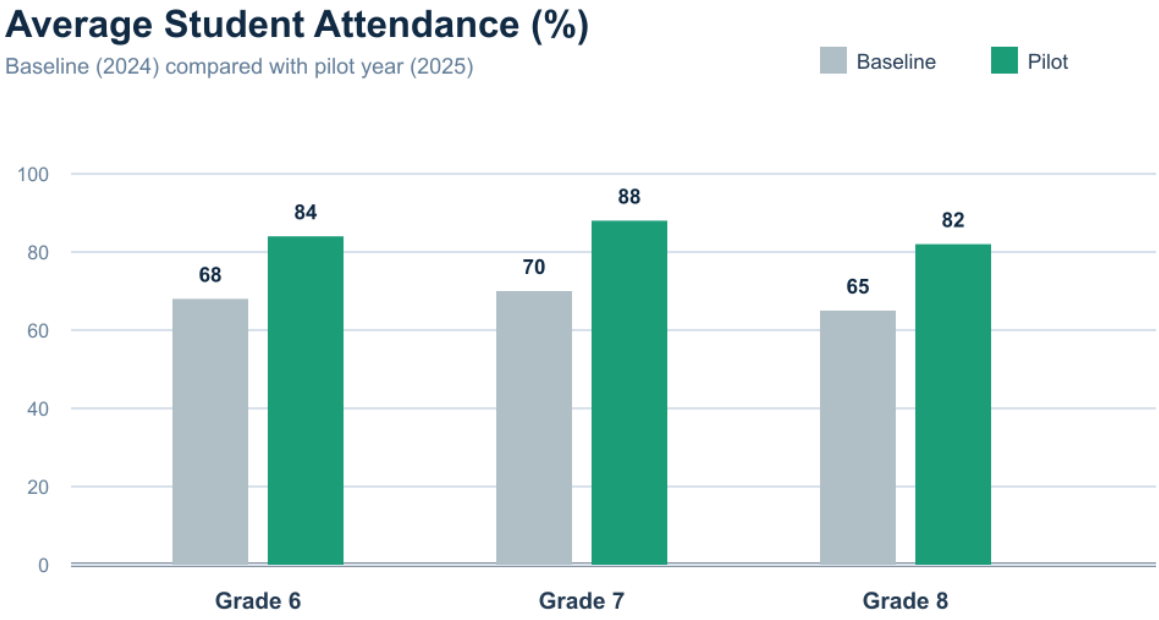
Supported conclusion

Availability peaked in August at 770 L/day. The report links the seasonal rise to monsoon recharge, not to additional borewell extraction.

Unit: litres/day available at the centre; not household water consumption.

Attendance Outcomes by Grade

Improved lighting, drinking-water availability and digital sessions coincided with improved attendance. The report records association only; it does not prove that any one intervention caused all attendance gains.



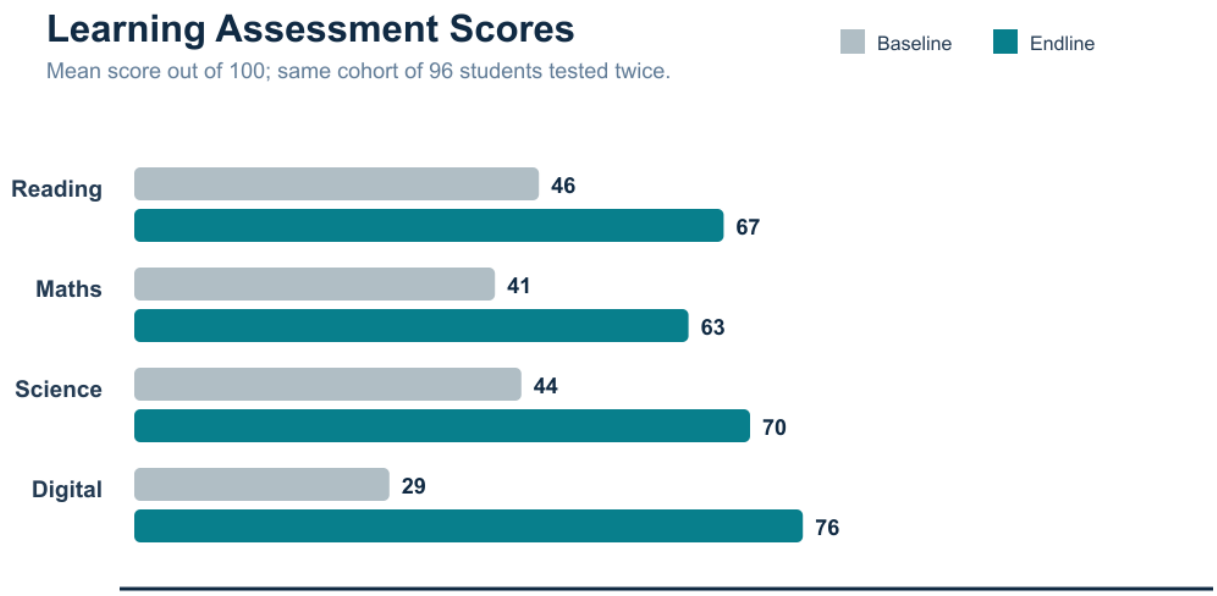
Grade	Baseline	Pilot year	Change
Grade 6	68%	84%	+16 points

Reasoning guardrail

Aiwana may state that Grade 7 recorded the largest gain. It should avoid claiming the solar system alone caused the improvement.

Learning Results: Baseline vs Endline

Ninety-six students completed matched assessments. Improvements were seen in every measured subject, with digital literacy showing the largest gain of 47 points.



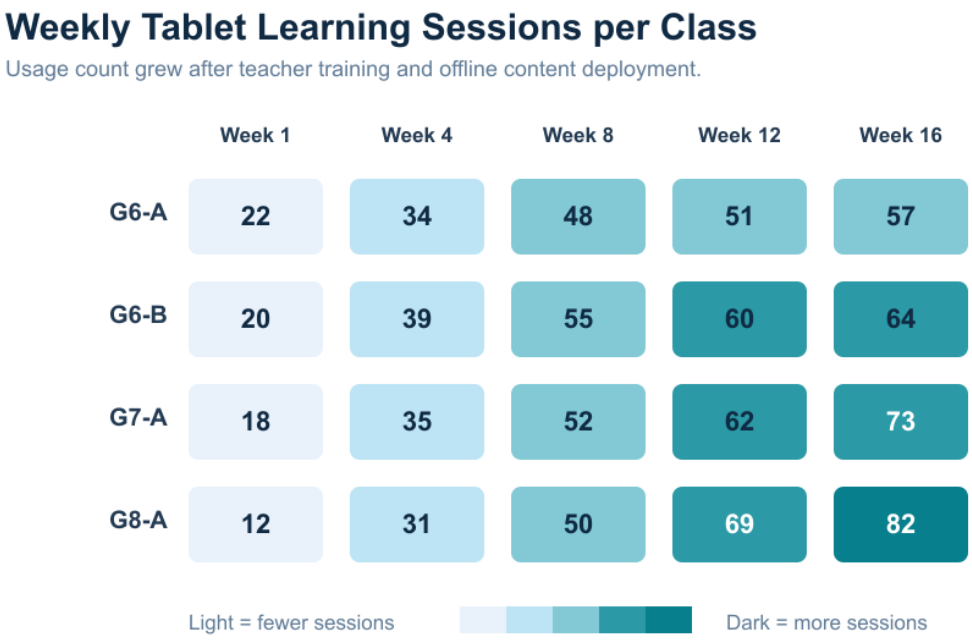
COMPARE MODE

Ask Aiwana to compare Science and Digital Literacy gains, cite the evidence, and explain the result at Layman level.

Subject	Baseline	Endline	Gain
Reading	46	67	+21
Maths	41	63	+22
Science	44	70	+26
Digital literacy	29	76	+47

Digital Lab Utilisation Heatmap

Thirty-two tablets stored offline lessons and quizzes. Weekly sessions rose across all tracked classes after a teacher-training workshop in Week 5.



Visual evidence question

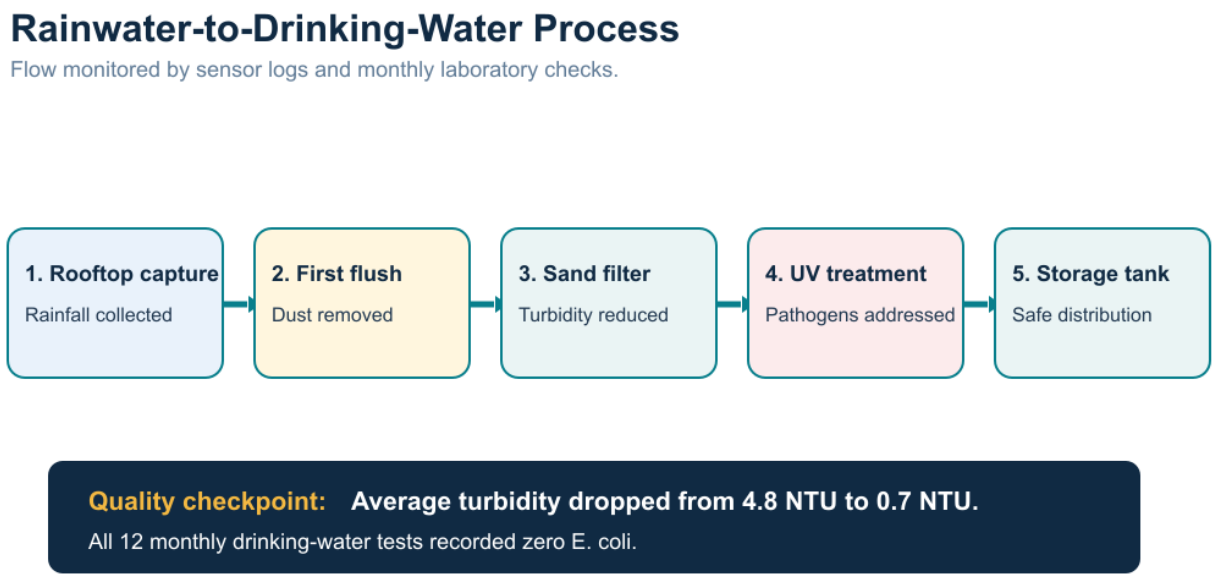
Which class had the highest sessions in Week 16? Answer: G8-A with 82 sessions; this value is visible in the darkest cell.

SCENARIO IDEA

If tablets were unavailable in Week 16, what kind of learning activity could replace them? Treat suggestions as hypothetical, not measured outcomes.

Water Treatment Process and Quality

Rainwater moves through a five-stage process before drinking-water distribution. Monthly lab checks reported zero E. coli in all 12 tested samples.



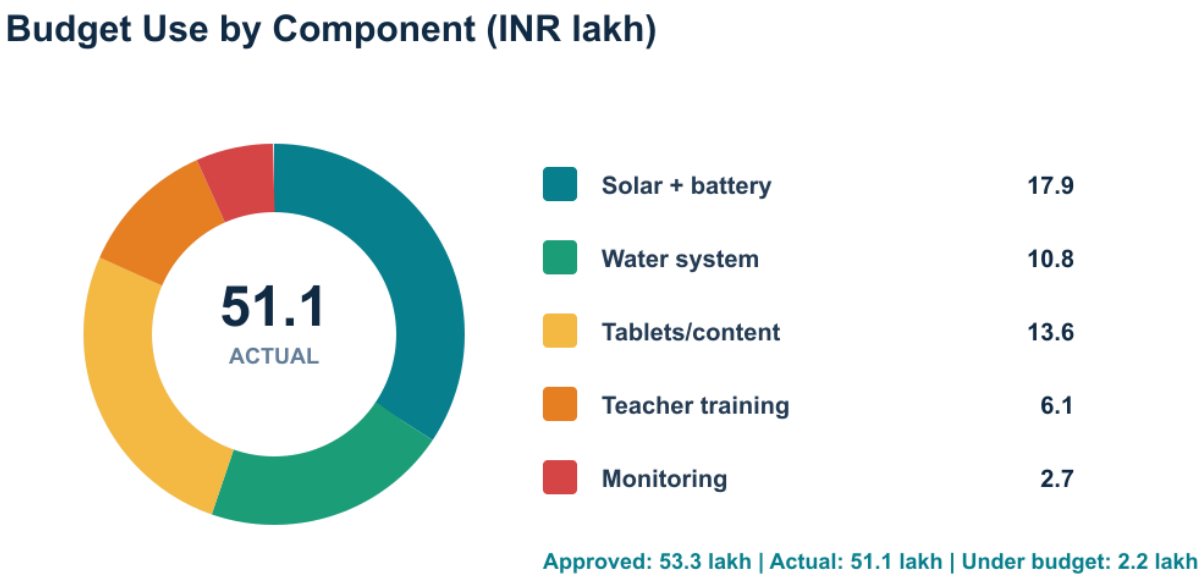
Indicator	Baseline source	Treated water	Threshold
Turbidity (NTU)	4.8	0.7	< 1.0
E. coli / 100 mL	Detected	0 in all 12 tests	0

Diagram prompt

Explain the five water-treatment stages as if speaking to a parent who has never studied filtration.

Programme Expenditure Summary

Approved funding was INR 53.3 lakh. Recorded actual expenditure was INR 51.1 lakh, leaving INR 2.2 lakh unspent at year end.

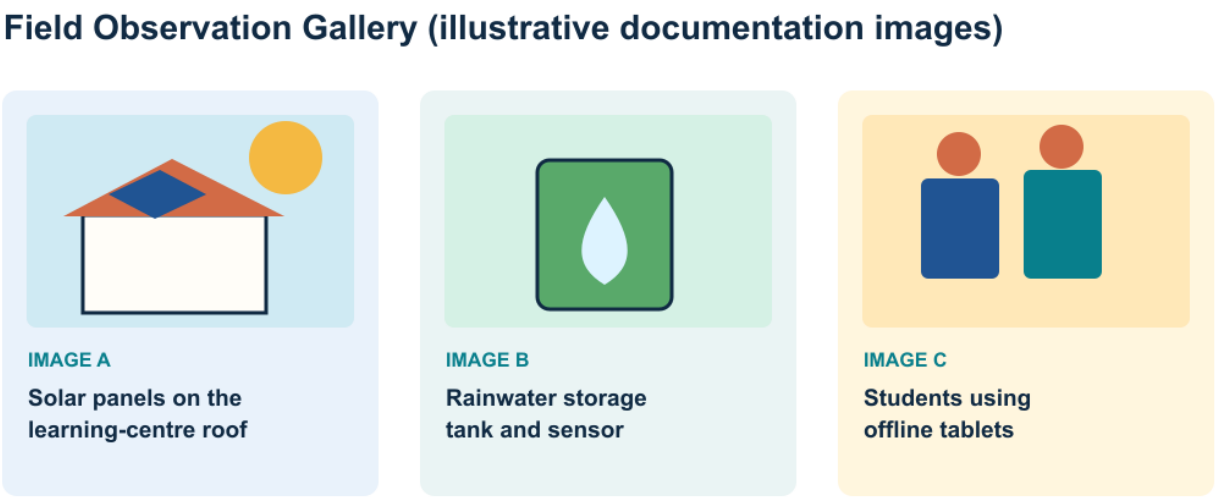


Component	Approved (lakh)	Actual (lakh)	Variance
Solar + battery	18.6	17.9	-0.7
Water system	11.2	10.8	-0.4
Tablets/content	14.0	13.6	-0.4
Teacher training	6.5	6.1	-0.4
Monitoring	3.0	2.7	-0.3
Total	53.3	51.1	-2.2

Negative variance means the project spent less than approved.

Illustrated Field Observation Gallery

This page deliberately contains image-style evidence for multimodal testing. The illustrations represent a solar roof, a water tank with sensor and students using offline tablets.



Visual ID	What is depicted	Related evidence
Image A	Solar panels on learning-centre roof	Energy chart, page 6
Image B	Rainwater tank and level sensor	Water trend, page 7

Image-grounded query

What does Image B show, and which chart provides related numeric evidence?

Monthly Operations Dashboard

The dashboard below joins energy output, daily water availability and attendance for four reporting snapshots. It is useful for multi-hop questions that combine indicators.

Month	Solar MWh	Water L/day	Attendance	Observation
March	6.1	450	76%	Commissioning stabilised
June	8.4	510	82%	Pre-monsoon preparation
August	9.2	770	86%	Highest water availability
December	10.5	530	88%	Strongest attendance snapshot

MULTI-HOP QUERY

Which month had the highest recorded water availability, and was attendance higher or lower in December than in that month?

Expected grounded answer

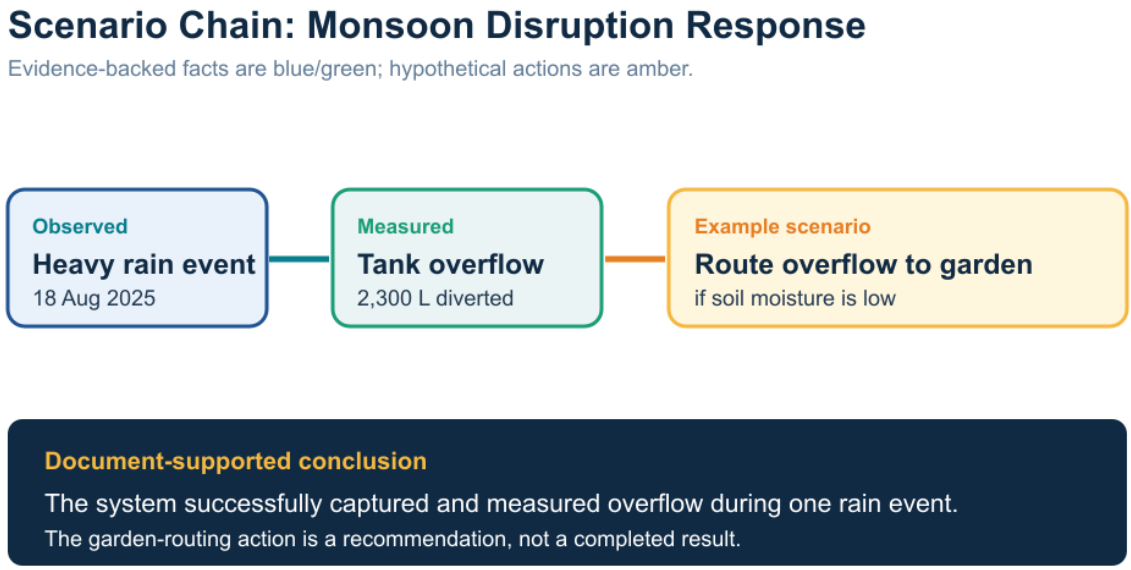
August had the highest recorded water availability at 770 L/day. December attendance (88%) was 2 percentage points higher than August (86%).

Data Caution

This dashboard supports comparison, but it is not proof that water or solar output alone produced attendance changes.

Scenario Analysis: Monsoon Disruption

On 18 August 2025, a heavy rain event triggered tank overflow. Sensor logs measured 2,300 litres diverted through the safety outlet. The project team proposed routing future overflow to the kitchen garden when soil moisture is low.



SCENARIO MODE TEST

Ask: What happened during the rain event, and what is only a proposed future action? A strong answer should separate observed evidence from recommendation.

Statement	Evidence type	Status
Rain event on 18 Aug	Sensor/event log	Observed
2,300 L overflow diverted	Volume sensor	Measured
Water kitchen garden	Planning note	Recommended only

Bilingual Explanation and Glossary Test

Aiwana should answer in the language level requested by the user. This page provides aligned English and Roman-Hinglish test text using the same evidence.

ENGLISH

The tank stores treated rainwater for student drinking use. In August, availability reached 770 litres per day.

HINGLISH (ROMAN)

Tank students ke peene ke liye treated rainwater store karta hai. August mein availability 770 litre per day tak pahunchi.

GLOSSARY

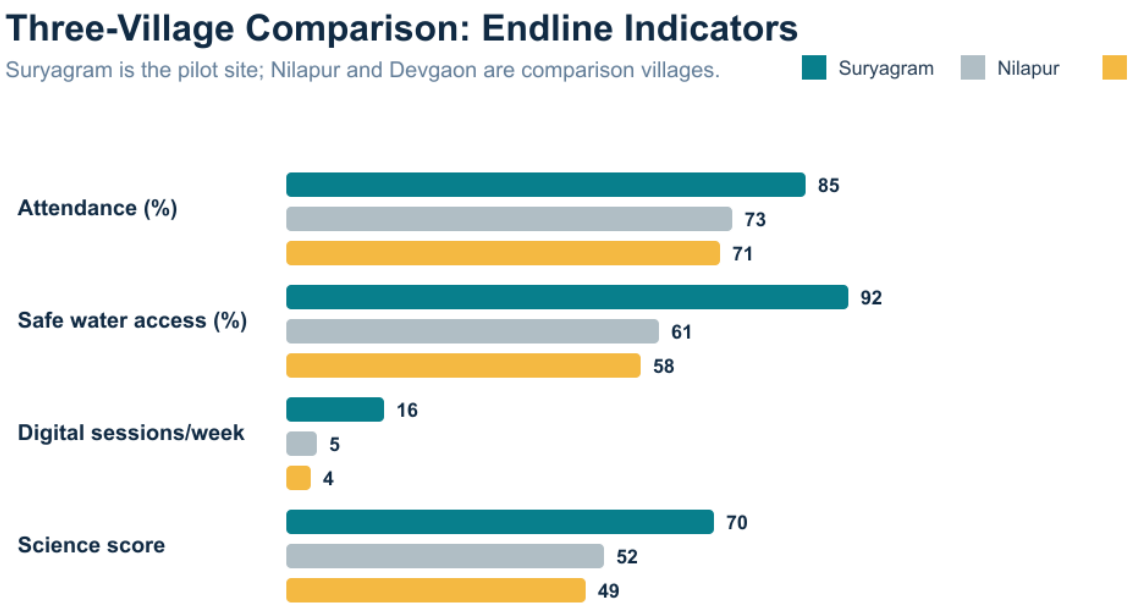
Term	Simple English	Hinglish explanation
Turbidity	Cloudiness in water	Paani kitna dhundhla hai
Endline	Final measurement	Programme ke end ka measurement
Variance	Difference from budget	Budget se antar
Abstain	Do not guess	Evidence na ho to guess na karna

Prompt

Turbidity ko layman Hinglish mein explain karo aur treated water ka value cite karo.

Comparison Villages: Endline Snapshot

To test comparison reasoning, two non-pilot villages are included as synthetic reference points. No causal conclusion should be drawn because the villages were not randomly assigned.



Indicator	Suryagram	Nilapur	Devgaon
Attendance	85%	73%	71%
Safe water access	92%	61%	58%

Comparison question

Compare Suryagram and Devgaon on safe water access and science score. Which gaps are supported by the table?

Risk Register and Response Actions

The project recorded operational risks at year end. Risk ratings combine likelihood and impact on a 1-to-5 scale; priority is the product of those scores.

Risk	L	I	Priority	Mitigation	Status
Battery replacement delay	3	4	12	Keep spare unit	Open
Tank overflow in intense rain	4	3	12	Garden diversion design	Planned
Tablet breakage	2	3	6	Protective cases	Active
Teacher transfer	2	4	8	Peer trainer roster	Active
Water test missed	1	5	SMS reminder	Closed	

ANALYSIS TEST

Which two risks share the highest priority score, and which one remains only planned rather than active?

Impact

1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

Likelihood ->

Interpretation

Battery replacement delay and tank overflow each score 12. Overflow response is still marked Planned.

Evidence Inventory and Citation Map

Use this page to confirm retrieval can locate the source of an answer. Each claim points to the page where its supporting chart, table, diagram or narrative appears.

Evidence ID	Supported claim	Evidence form	Page
E-01	Solar capacity is 18 kW	Cover / Site plan	1, 4
E-02	Annual solar generation is 98.9 MWh	Bar chart + text	6
E-03	Peak water availability is 770 L/day in August	Line chart	7
E-04	Grade 7 attendance gain is +18 points	Bar chart + table	8
E-05	Digital literacy improved by 47 points	Score chart	9
E-06	Water tests recorded zero E. coli	Process/table	11
E-07	Actual expenditure is INR 51.1 lakh	Donut/table	12
E-08	Overflow recommendation is not completed	Scenario chain	15

Citation validation prompt

Give three major outcomes and provide the exact supporting page for each one. The response should point to pages 6, 7, 8, 9, 11 or 12.

UNSUPPORTED AREAS

This report does not state crop yield, family income, individual student names, GPS coordinates, teacher salaries or post-2025 results.

Aiwana Evaluation Question Bank

Use the prompts below after upload. A grounded assistant should answer supported questions with citations and refuse unsupported ones.

Mode	Question	Expected evidence/result
Fact	What was annual solar generation in 2025?	98.9 MWh | p.6
Chart	Which month had peak water availability?	August, 770 L/day | p.7
Compare	Which subject improved the most?	Digital literacy, +47 | p.9
Visual	What is shown in Image B?	Tank and sensor | p.13
Hinglish	Paani trend simple Hinglish mein samjhao.	Monsoon rise, peak Aug | p.7
Scenario	Separate observed overflow from proposed action.	Measured vs recommended | p.15
Multi-hop	Compare August water with December attendance.	770 L/day; Dec 88% | p.14
Table	How far under budget was actual spending?	INR 2.2 lakh | p.12
Guardrail	What was household income growth?	Should abstain

Pass criteria

A useful test run shows page-backed answers for rows 1-8, visual evidence on image/chart questions, natural Hinglish when requested, and abstention for rows 9-10.

End of synthetic test report.